Interview Prep: SQL Developer L1

Simple Logic IT Pvt Ltd | Vikhroli, Mumbai | WFO | 1-3 years | Prepared for Hritik Maurya

How to use this: Answers are in simple English so you can speak them naturally. Do not memorize word-by-word; understand the point and say it in your own style. Anything in [square brackets] is for you to fill with your real details.

PART 1: HR and Job Switch Questions (Notice Period, Payroll, etc.)

Q1. Tell me about yourself.

I am Hritik Maurya from Mumbai. I completed my BSc IT from SM Shetty College, Mumbai University in 2024. Since May 2025, I am working as an SQL Developer at Uni Design Jewellery, where I write and optimize SQL queries and stored procedures and create reports for business decisions. Before that, I did internships in MERN stack, and I have certifications in MySQL, JavaScript, React and C++. I enjoy working with data, and now I want to go deeper into query optimization and performance tuning. That is why this role at Simple Logic interests me.

Tip: *Keep it under 1 minute. Present, then past, then why this role.*

Q2. Why do you want to leave your current job?

I am thankful to my current company, I learned a lot about real SQL work there. But now I want to grow in query optimization and performance tuning on bigger databases, and work towards a DBA-type role. This role matches that direction, so this is a career growth decision, not any problem with my company.

Tip: *Never say anything negative about your current company, manager or salary.*

Q3. What is your notice period? Can you join immediately?

My notice period is 1 month as per my company policy. I am on the company payroll, so I have to follow the process. I am requesting my manager and HR for an early release, and I will inform you about the exact joining date as soon as I get the confirmation. In the worst case, I can join in 30 days.

Tip: *Be honest. Do not promise 'immediate joining' unless your HR has confirmed it in writing. A relieving letter is important for your background verification at the new company. If the offer is confirmed, then request early release.*

Q4. You are on payroll, so will your company release you early? Can you do a notice period buyout?

Yes, I am on payroll. My appointment letter says 1 month notice. I will discuss with my HR about early release or a buyout, if the policy allows it. I will share the written confirmation with you once I have it.

Tip: *Before the interview, check your appointment letter for: notice period, buyout clause, and any bond or penalty.*

Q5. Why are you in such a hurry to switch?

I am not in a hurry, I am serious about this opportunity. The role is exactly about SQL scripting and performance tuning, which is the field I want to build my career in. I want to start as early as possible, but I will complete my handover properly and leave my current team on good terms.

Q6. The service agreement is 2 years. Are you okay with that?

Yes, I understand. I am looking for a long-term place to learn and grow, so 2 years is not a problem for me. I would like to read the agreement once fully, especially the exit terms, and then I am comfortable to go ahead.

Tip: *Ask politely about the bond amount and exit conditions, if any. Get it in writing.*

Q7. This is Work From Office at Vikhroli. Is that okay? How will you travel?

Yes, I am comfortable with work from office. I stay in Bhandup, and Vikhroli is a short local train ride on the Central line, so travel is easy for me.

Q8. What are your salary expectations?

My current CTC is [your current CTC]. Based on my skills and the market, I am expecting around [your expected CTC]. But I am flexible, and I mainly care about a good learning opportunity and a fair offer as per company standards.

> ***Tip:*** *Give a range, not one fixed number. Ask for a 25-30% hike as a realistic target.*

Q9. Why should we hire you?

I have more than a year of hands-on SQL experience in a real business. I write queries and stored procedures daily, and I build reports. I learn new tools fast, I am serious about performance tuning, and I am ready to work as a Performance Tuning DBA. I am also a team player and I take responsibility for my work.

Q10. What are your strengths and weaknesses?

Strength: I learn quickly and I like solving problems step by step. Weakness: sometimes I spend too much time on perfecting one query. Now I set a time limit, and I ask my senior for a review if I am stuck.

Q11. Where do you see yourself in 3 to 5 years?

I want to become a strong database professional, first as a good SQL developer, then a Performance Tuning DBA, who can handle large databases and help the team solve difficult performance problems.

Q12. Are you attending other interviews or holding other offers?

I have applied at a few places, but this role matches my career goal the best. If I get selected, I will give my full commitment to Simple Logic.

> ***Tip:*** *Keep it simple. Do not lie about offers, and do not create pressure.*

PART 2: SQL Basics

Q13. What is SQL? What are DDL, DML, DCL and TCL?

SQL (Structured Query Language) is used to store, read, update and manage data in a relational database.

- **DDL** (structure): CREATE, ALTER, DROP, TRUNCATE
- **DML** (data): INSERT, UPDATE, DELETE, SELECT
- **DCL** (permissions): GRANT, REVOKE
- **TCL** (transactions): COMMIT, ROLLBACK, SAVEPOINT

Q14. Difference between DELETE, TRUNCATE and DROP?

- **DELETE**: removes selected rows (can use WHERE), can be rolled back, slower, fires triggers.
- **TRUNCATE**: removes all rows quickly, resets identity, no WHERE.
- **DROP**: removes the whole table (structure + data).

Q15. Primary key vs Foreign key vs Unique key?

- **Primary key**: uniquely identifies each row, no NULL, only one per table.
- **Foreign key**: links to the primary key of another table, keeps data consistent.
- **Unique key**: values must be unique, can allow NULL, many per table.

Q16. What are the types of JOINs?

- **INNER JOIN**: only matching rows from both tables.
- **LEFT JOIN**: all rows from left + matching from right.
- **RIGHT JOIN**: all rows from right + matching from left.
- **FULL JOIN**: all rows from both tables.
- **CROSS JOIN**: every row with every row (Cartesian product).
- **SELF JOIN**: a table joined with itself (e.g. employee and manager).

Q17. WHERE vs HAVING?

WHERE filters rows before grouping. HAVING filters groups after GROUP BY (mostly with aggregate functions).

```
SELECT dept_id, COUNT(*) AS emp_count
FROM employees
WHERE status = 'Active'
GROUP BY dept_id
HAVING COUNT(*) > 5;
```

Q18. UNION vs UNION ALL?

UNION removes duplicates (slower because it sorts/compares). UNION ALL keeps duplicates (faster). If I know there are no duplicates, I prefer UNION ALL for performance.

Q19. What is normalization?

It is organizing data to reduce duplication and keep it consistent. 1NF: atomic values, no repeating groups. 2NF: 1NF + no partial dependency. 3NF: 2NF + no transitive dependency. Sometimes we denormalize on purpose for reporting speed.

Q20. What is an index? Clustered vs Non-clustered?

An index is like a book index, it helps the database find rows faster without scanning the whole table.

- **Clustered**: decides the physical order of data in the table. Only one per table.
- **Non-clustered**: separate structure with pointers to the data. Many per table.

Q21. View vs Stored Procedure vs Function vs Trigger?

- **View**: saved SELECT query, works like a virtual table.
- **Stored procedure**: saved set of SQL statements, can take parameters, do inserts/updates, handle transactions.
- **Function**: returns a value or table, can be used inside a SELECT.
- **Trigger**: runs automatically on INSERT/UPDATE/DELETE.

Q22. What is a CTE? Subquery vs JOIN?

A CTE (Common Table Expression) is a temporary named result set using WITH. It makes complex queries easy to read and supports recursion. A subquery is a query inside a query. In many cases a JOIN is faster and cleaner, but modern optimizers often treat both similarly, so I check the execution plan.

Q23. What are ROW_NUMBER, RANK and DENSE_RANK?

All are window functions used for ranking. ROW_NUMBER gives a unique number to each row (1,2,3,4). RANK gives the same rank for ties and skips the next number (1,2,2,4). DENSE_RANK gives the same rank for ties and does not skip (1,2,2,3).

Q24. How do you handle NULL values?

Use IS NULL / IS NOT NULL (not = NULL). Use ISNULL() or COALESCE() to replace NULL with a default value. Remember that NULL in a calculation gives NULL, and COUNT(column) ignores NULLs while COUNT(*) does not.

Q25. What is a transaction and what is ACID?

A transaction is a group of statements that run as one unit. ACID: **Atomicity** (all or nothing), **Consistency** (data stays valid), **Isolation** (transactions do not disturb each other), **Durability** (once committed, data is saved permanently).

PART 3: T-SQL / PL-SQL (Main Requirement in the JD)

Q26. Do you have experience in T-SQL or PL-SQL?

I work with SQL daily in my current role: queries, stored procedures and reports. My hands-on database is [MySQL / SQL Server / Oracle - write your real one]. The core concepts (joins, procedures, functions, transactions, error handling) are the same in T-SQL and PL-SQL, and I am learning the syntax differences quickly.

> ***Tip:*** *Be honest about which one you actually used. Interviewers respect honesty, and they will test you. If you have not used T-SQL, spend 2 days on SQL Server Express (free) before the interview.*

Q27. T-SQL vs PL-SQL?

- **T-SQL** (Transact-SQL): Microsoft SQL Server's extension of SQL.
- **PL-SQL**: Oracle's procedural extension of SQL.
- Both add variables, IF/ELSE, loops, procedures, functions, triggers and error handling.
- Example: T-SQL uses TOP, ISNULL, TRY...CATCH; Oracle uses ROWNUM / FETCH FIRST, NVL, EXCEPTION block.

Q28. Write a simple stored procedure with a parameter.

```
CREATE PROCEDURE GetOrdersByCustomer
    @CustomerId INT
AS
BEGIN
    SET NOCOUNT ON;
    SELECT OrderId, OrderDate, Amount
    FROM Orders
    WHERE CustomerId = @CustomerId
    ORDER BY OrderDate DESC;
END
```

Q29. How do you handle errors in T-SQL?

Using TRY...CATCH. Inside CATCH I can rollback the transaction and read ERROR_MESSAGE().

```
BEGIN TRY
    BEGIN TRAN;
    UPDATE Accounts SET Balance = Balance - 500 WHERE Id = 1;
    UPDATE Accounts SET Balance = Balance + 500 WHERE Id = 2;
    COMMIT;
END TRY
BEGIN CATCH
    ROLLBACK;
    SELECT ERROR_MESSAGE() AS ErrorMsg;
END CATCH
```

Q30. Temp table vs Table variable vs CTE?

- **Temp table (#temp)**: stored in tempdb, can have indexes and statistics, good for big data.
- **Table variable (@t)**: lighter, good for small data, limited statistics.
- **CTE**: not stored, just a named query used in the next statement.

Q31. What is a cursor? Should we use it?

A cursor processes rows one by one. It is slow for large data. I prefer set-based queries (JOIN, window functions, UPDATE with JOIN). I use a cursor only when there is no set-based option.

PART 4: Query Rewriting and Performance Tuning (Key Focus)

Q32. A query is running slow. How will you find and fix the problem?

I follow steps:

- 1. Check the **execution plan** to see where the cost is (scans, key lookups, sorts, joins).
- 2. Check if proper **indexes** exist on JOIN, WHERE and ORDER BY columns.
- 3. **Rewrite the query**: select only needed columns, remove functions on columns, avoid cursors.
- 4. Check **statistics** are updated.
- 5. Test again and compare time and reads (SET STATISTICS IO, TIME ON).

Q33. What is a SARGable query?

SARGable means the query can use an index seek. If I wrap a column in a function, the index cannot be used properly, so I rewrite it.

```
-- Not SARGable (index not used well)
SELECT * FROM Orders WHERE YEAR(OrderDate) = 2025;

-- SARGable (rewritten)
SELECT OrderId, Amount FROM Orders
WHERE OrderDate >= '2025-01-01' AND OrderDate < '2026-01-01';
```

Q34. Give examples of query rewriting for better performance.

- Replace **SELECT *** with only the required columns.
- Avoid a leading wildcard: LIKE '%abc' cannot use an index, LIKE 'abc%' can.
- Use **EXISTS** instead of IN for large subqueries.
- Replace cursors/loops with set-based queries.
- Avoid functions on indexed columns in WHERE.
- Filter early and reduce data before joining.

Q35. What is an execution plan? What do you look for?

It shows how the database runs the query. I look for: Table Scan / Index Scan (bad on big tables) vs Index Seek (good), Key Lookups, expensive Sort or Hash operators, big difference between estimated and actual rows, and missing index suggestions.

Q36. Index Seek vs Index Scan?

Seek jumps directly to the matching rows using the index (fast). Scan reads the whole index or table (slower on big data).

Q37. What is a covering index?

An index that contains all the columns needed by a query (using INCLUDE columns), so the database does not need to go back to the table (no key lookup).

Q38. Can too many indexes be a problem?

Yes. Indexes speed up reads but slow down INSERT, UPDATE and DELETE, and use more storage. So I create only useful indexes and remove unused ones.

Q39. What are statistics and why do they matter?

Statistics tell the optimizer how the data is distributed. If they are outdated, the optimizer picks a bad plan and the query becomes slow. Updating statistics regularly helps.

Q40. What is blocking and deadlock?

Blocking: one transaction holds a lock and another waits for it. **Deadlock**: two transactions wait for each other forever, so the database kills one. To reduce: keep transactions short, access tables in the same order, use proper indexes.

Q41. Have you done any performance tuning in your job?

Yes. In my current role I [describe one real example: e.g., a report query that took X minutes; I checked the plan, added an index / rewrote a subquery to a join / removed a function from WHERE, and it came down to Y seconds].

Tip: *Prepare ONE real, small example from your work. This is the most important answer for this role. Use this format: Problem - What I checked - What I changed - Result.*

Q42. The JD says 'willing to work as a Performance Tuning DBA'. Are you okay with that?

Yes, I am very interested. I already enjoy optimizing queries and reading execution plans. I know it needs learning about indexes, statistics, blocking and monitoring, and I am ready to learn and grow in that direction.

PART 5: Practical Query Questions (Very Common)

Q43. Find the second highest salary.

```
-- Using DENSE_RANK (best, works for Nth)
SELECT salary FROM (
    SELECT salary, DENSE_RANK() OVER (ORDER BY salary DESC) AS rnk
    FROM employees
) t
WHERE rnk = 2;

-- Simple way
SELECT MAX(salary) FROM employees
WHERE salary < (SELECT MAX(salary) FROM employees);
```

Q44. Find duplicate records.

```
SELECT email, COUNT(*) AS cnt
FROM users
GROUP BY email
HAVING COUNT(*) > 1;
```

Q45. Delete duplicate rows but keep one.

```
WITH cte AS (
    SELECT *, ROW_NUMBER() OVER (PARTITION BY email ORDER BY id) AS rn
    FROM users
)
DELETE FROM cte WHERE rn > 1;   -- SQL Server syntax
```

Q46. Find customers who have not placed any order.

```
SELECT c.CustomerId, c.Name
FROM Customers c
LEFT JOIN Orders o ON o.CustomerId = c.CustomerId
WHERE o.OrderId IS NULL;
```

Q47. Top 3 highest paid employees in each department.

```
SELECT * FROM (
    SELECT *, DENSE_RANK() OVER (PARTITION BY dept_id ORDER BY salary DESC) AS rnk
    FROM employees
) t
WHERE rnk <= 3;
```

Q48. Employees who earn more than their manager (self join).

```
SELECT e.name
FROM employees e
JOIN employees m ON e.manager_id = m.id
WHERE e.salary > m.salary;
```

Q49. Show running total of sales by date.

```
SELECT SaleDate, Amount,
       SUM(Amount) OVER (ORDER BY SaleDate) AS RunningTotal
FROM Sales;
```

PART 6: Questions YOU Can Ask the Interviewer

- What database platform do you mainly work on: SQL Server or Oracle? What size are the databases?
- What will be my day-to-day work in the first 3 months? Is there any training for Performance Tuning DBA work?
- How is the team structured, and who will I report to?
- What are the working hours, and what is the interview process after this round?
- What is the expected joining timeline? (Good place to mention your notice period and early release discussion.)

Quick Checklist Before the Interview

- Read your appointment letter: notice period, buyout option, bond or penalty.
- Prepare ONE real performance tuning example from your current work.
- Practice the 7 queries in Part 5 on your own laptop until you can write them without looking.
- Know your current CTC, expected CTC and exact notice period date.
- Do not talk badly about your current company. Do not promise 'immediate joining' without HR confirmation.
- Keep your resume open and be ready to explain every skill and project listed on it.
- Carry ID proof, latest payslip and offer/appointment letter if asked for documents.

All the best, Hritik!